SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

ENG201

Timed Online Assignment – January Semester 2021

Linear Systems Analysis & Design

Monday, 17 May 2021 **1:00 pm – 3:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **FIVE (5)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOACourse Code, Your Name and Student PI No. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT space.
   Example, TOAENG201_RaphaelLee_T1923161 (omit D/O, S/O).

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6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a Microsoft Word file saved in .doc or .docx format. All answers are to be typed in Times New Roman font, size 12 and single spacing. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. Flowcharts and graphs may be scanned or photographed and embedded in the Word file in the right sequence provided it does not exceed the file size limit of 500MB. All uploaded handwritten answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

7. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

8. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

Answer all questions. (Total 100 marks)

Question 1

Analyze the following signals and determine their period.

(a) (i) $x_1[n] = \cos(7\pi n)$

(3 marks)

(ii) $x_2[n] = \sin^2(3\pi n)$

(5 marks)

(iii) $x_3[n] = \cos(7\pi n) - 2\sin(3\pi n)$

(5 marks)

(b) Sketch the waveform of the following signal $y(t)$

$$y(t) = r(t+3) + u(t+2) - r(t+2) + r(t+1) - 2r(t) + r(t-1) - u(t-2) - r(t-2) + r(t-3)$$

(6 marks)

(c) Given $z(t) = y(-2t+1)$. Describe the **THREE (3)** changes in the waveform $z(t)$ compared with $y(t)$ due to the different signal operations.

(6 marks)

Question 2

(a) Consider the system in Figure Q2(a) with three inter-connected subsystems whose impulse responses are $h_1(t), h_2(t),$ and $h_3(t)$, respectively.

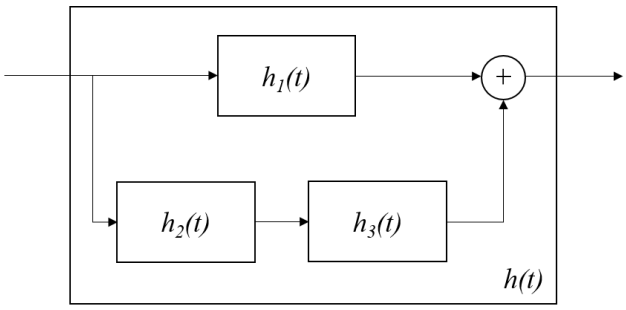

Figure Q2(a)

(i) Describe the impulse response $h(t)$ of the system using the impulse responses of each subsystem.

(5 marks)

(ii) If $h_1(t) = e^{-|t|}$, then calculate the step response of the subsystem $h_1(t)$.

(10 marks)

(b) Determine the homogeneous solution and particular solution of the following system

$$y[n] + \frac{1}{9}y[n-2] = x[n] + 3x[n-2],$$

where $x[n] = \left(\frac{1}{3}\right)^n$.

(10 marks)

Question 3

Calculate the DTFS coefficients of the following discrete-time periodic signals:

(a)

$$x_1[n] = 3\sin\left(\frac{12}{37}\pi n\right) + 2\cos\left(\frac{10}{37}\pi n\right)$$

(8 marks)

(b) Solve for the Fourier series coefficients corresponding to the continuous-time periodic signals in Figure Q3(b).

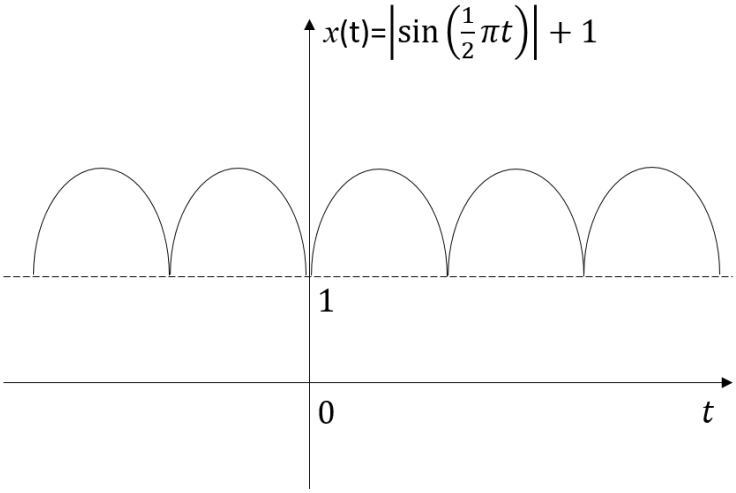

Figure Q3(b)

(9 marks)

(c) Find the Laplace Transform of the following signals:

(i) $x(t) = \frac{d}{dt}(e^{-3t}u(t))$

(4 marks)

(ii) $x(t) = t \times \cos(101\pi t)u(t)$

(4 marks)

Question 4

Given a LTI system

$$\frac{d^2}{dt^2}y(t) + 3\frac{d}{dt}y(t) + 11y(t) = \frac{d}{dt}x(t)$$

With

$$y(0-) = 1, \frac{d}{dt}y(t)|_{t=0-} = 0, x(t) = u(t)$$

(a) Design the transfer function corresponding to the given system.

(9 marks)

(b) Discuss the stability of the given system.

(6 marks)

(c) Determine the forced response corresponding to the given system.

(10 marks)

----- END OF PAPER -----